

Measured, Not Asked: Free Saliency Signals for Consolidation and Routing at Small Scale

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Abstract

An agent that moves knowledge from its context into its weights, or decides which part of a long context to read, needs a saliency signal. Three such signals come free, with no reward model or judge: the keep/drop decision a context compactor already makes, the divergence between a model’s predictions with and without a document in context, and the region a model declares it needs to attend to. We test all three on Qwen2.5 models from 0.5B to 7B with a matched control beside every signal-strength figure, and report two findings. **First, measured beats asked.** Signals the model is asked to state fail or are weak: asked to list spans worth keeping, 0.5B and 1.5B models return the prefix 0, 1, 2, ... on every event; asked to declare which of eight regions holds an answer, the 1.5B model does no better than naming a fixed region. The same models’ measured signals work: reading the keep decision off the logits gives a 2.56x saliency lift, and scoring each region by the log-odds of a relevance question routes at 0.33–0.42 hit rate against 0.125 chance, beating the stated declaration at 0.5B, 1.5B and 7B (by 0.19 at 7B, $t(2)=24$) and late-layer attention probing at 0.5B and 7B. **Second, a salient signal is not necessarily a good training target.** Consolidating on the compactor’s keep decision, a $+15.85\sigma$ saliency signal, recovers fewer evicted facts than consolidating on uniformly sampled spans on all five synthetic corpora (1.7% vs 7.6%, paired $t(4)=3.10$), because the compactor has systematic blind spots; and gating distillation on the context gap performs no better than random token selection under identical masking (5 seeds, $t(4)=0.74$). Neither free signal beats uncured coverage as a training target. Matched controls caught eight artifacts in our own pipeline, six of which had inflated a result in our favour.

1 Introduction

Context is expensive and transient; weights are cheap to read and permanent. A long-running agent therefore wants to move what matters from one to the other, and to read only the part of its context a question needs. In both cases the training or routing step is routine — a LoRA pass, a sparse KV read — and the hard part is deciding *what* matters.

Prior work pays for that decision. Nightly consolidation (Dennis et al., 2026) writes reflections with a second generation pass; SCoL (Wang et al., 2026) learns where to write with meta-RL; CompactionRL (Li et al., 2026) trains the compactor with RL. We ask what is already free, and find three candidates:

- **The compaction decision.** Every time an agent compacts its context, it decides which spans survive verbatim. That is a saliency label, emitted at no extra cost.
- **The context gap.** Run a frozen model on the same continuation with and without a skill document in context. The per-token KL divergence between the two is the document’s causal contribution, from two forward passes.
- **The attention declaration.** Declarative Attention (Ho et al., 2026) has a model state which region of its context it needs, and skips the rest of the KV-cache read. The declaration is also a saliency label.

We test each as a training or routing signal on models a single GPU can run, 0.5B to 7B. The results sort along two axes. The first is how the signal is elicited: *measured* from the model’s logits or activations, or *asked* of it as generated text. This decides whether a signal exists at small scale. The second is whether a signal that exists is the right thing to train on. This decides whether it helps.

Contributions. This is a measurement paper: the methods are hypotheses under test, and two of the three consolidation hypotheses come back negative.

1. **Measured beats asked** (§4). On three independent mechanisms, a signal the model is asked to state fails or is weak, while the same model’s measured signal works. A generated span list is exactly random; the same keep decision read off the logits gives a 2.56x lift. A generated attention declaration is at a fixed-slot control at 1.5B and carries under half the content signal of a logit read at 0.5B and 7B.
2. **Salient is not trainable** (§5). Two free signals that clear matched controls do not beat uniform coverage as training targets. Compaction-gated consolidation is significantly worse than uniform replay across five corpora; context-gap gating is indistinguishable from random token selection. They fail for different reasons — coverage and neutrality.
3. **A logit-read substitute for declarative attention below the scale where declaring works** (§4.2): a cheap relevance query that routes at about 3x chance at every scale tested, with a content-shuffle control that separates content-driven from position-driven routing.
4. **A matched control beside every signal-strength figure** (§6), which caught eight artifacts in our own pipeline.

2 Related work

Consolidation from context. Beyond Inference-Only Deployment (Dennis et al., 2026) shows that periodic LoRA consolidation beats cascading compaction, and establishes median rather than mean per-token cross-entropy as the metric that tracks judged accuracy ($r = +0.99$ against -0.51). Its targets are reflected facts written by a second generation pass; we consolidate on the compactor’s raw decision instead, which removes that pass and reflection quality as a confound. Their recipe is our baseline. SCoL (Wang et al., 2026) learns *where* to write consolidated knowledge — the complementary axis to our *what*. What to Keep, What to Forget (Colaco & Lahjouji, 2026) frames compaction as rate-distortion and stops at the compactor; we consume its decisions as training signal. CompactionRL (Li et al., 2026) trains the compactor with RL; we hold it frozen and train the backbone, and the two compose. Compression-Aware Abstention (Khodabandehlou & Krishnamachari, 2026) trains a model on KV-compression masks to refuse when compression removed the answer evidence; we reproduce its direction at 0.5B (§5.1).

Importance-gated distillation and token selection. Skill-to-LoRA (Zhang & Qi, 2026) distills each skill into its own adapter by uniform self-distillation, and is our baseline in §5.2. ThinkSwitch (Saini & Pandey, 2026) distills context into LoRA adapters with weight interpolation, also ungated. PEAM (Guo et al., 2026) internalises an embodied agent’s episodic experience into its parameters by contrasting trajectories — the nearest neighbour to our gate, on different objects with a different signal. Titans (Behrouz et al., 2024) uses the gradient of an associative-memory loss as surprise at test time; our context gap is a surprise signal in the same spirit, computed as a divergence between two forward passes and applied offline. Rho-1 (Lin et al., 2024) and TIP (Xu et al., 2026) report the opposite of our §5.2 result in their settings: selective loss on scored tokens helps pretraining (Lin et al., 2024), and training on under 10% of tokens, chosen by teacher–student divergence and entropy, nearly matches full-token on-policy distillation (Xu et al., 2026). Our design isolates the ranking from the masking with a random-subset control at identical masking. In our setting the subsetting carries the benefit and the ranking adds nothing; whether that decomposition holds in theirs is open.

Attention routing. Declarative Attention (Ho et al., 2026) has a 27–31B model declare `<global>`, `<focus>` or `<local>` in its chain of thought and skips most of the KV-cache read accordingly: 52.0% and 31.1% fewer attended tokens at -1.27 and -2.75 points of accuracy, zero-shot, with accuracy losses that “shrink with model scale”. We test the declaration from below, at the scales a single GPU runs. Sentinel (Zhang et al.,

2025) and the retrieval-head line (Wu et al., 2024) route by probing attention instead of asking; we run that route on the same models, layouts and probes as the other two.

3 Setup and measurement protocol

Models and compute. Qwen2.5-Instruct (Qwen Team, 2024) at 0.5B, 1.5B and 7B; LoRA (Hu et al., 2021) rank 16, $\alpha = 32$, on all attention and MLP projections. 0.5B and 1.5B run in fp16; 7B runs in nf4 4-bit quantisation to fit a single 16 GB T4, which is all the compute used.

Saliency lift. For a signal that marks spans, lift is the keep rate of fact spans over the keep rate of filler spans. Every lift is reported against a **positional control** — what keeping the first N spans scores on the same data — and with its σ over chance and over that control.

Statistics. Single-run comparisons use a binomial σ . Multi-seed comparisons use a paired t across seeds against the two-sided 5% critical value for $k-1$ degrees of freedom (4.30 at three seeds, 2.78 at five). We call a margin significant only if it clears that value; a difference that does not is reported as a tie, not a trend.

Content shuffle (§4.2). For routing signals, a raw hit rate cannot distinguish tracking content from favouring a position. We permute which content sits in which region and re-elicit. **content_dependence** is the shuffled hit rate above chance; **slot_stable_rate** is how often the choice stays put when the content moves (near $1/K$ is content-driven, near 1 position-driven). A best-constant control — always naming the most common gold region — exposes a model that names one region regardless.

4 Finding 1: measured beats asked

4.1 The compaction decision

The compactor segments the evictable prefix of a trajectory into sentence spans and keeps $\text{ceil}(\rho \cdot n)$ of them, $\rho = 0.25$. We elicit the keep decision two ways: **generate**, where the model lists span indices to keep; and **read**, where the model is asked of each span whether to keep it and we score $\log p(\text{Yes}) - \log p(\text{No})$ off the logits, keeping the top $\rho \cdot n$. Spans are shuffled before presentation so position cannot masquerade as salience.

Elicitation	Backbone	Lift	Verdict
generated index list	Qwen2.5-0.5B / 1.5B	0.53x	below positional control
logit read	SmolLM2-360M	2.17x	clears
logit read	Qwen2.5-0.5B	0.42x	fails
logit read	Qwen2.5-1.5B	2.56x	clears

Asked to list spans, both Qwen models return the contiguous prefix 0, 1, 2, ... on 100% of events regardless of content — under shuffling, exactly random selection. Read off the logits, the 1.5B model’s decision is a strong signal. On the corrected synthetic benchmark used for §5.1 (48 trajectories, 97 compaction events, 6,244 spans), it keeps fact spans at 0.663 and filler at 0.236: a 2.81x lift, **+15.85 σ over chance and +8.40 σ over the positional control**, with zero fallbacks or empty keeps. Per fact type it is uneven: **db_engine**, **owner** and **version_pin** are kept on 100% of events, **config_flag** on 3.4% and **auth_header** on 0%. §5.1 shows why that matters.

Measuring is necessary but not sufficient for a signal to exist: the 0.5B logit read fails (0.42x) where SmolLM2-360M clears (2.17x) on identical data and code. Whether a model carries a usable signal is a property of its behaviour on the probe, not of its size.

4.2 The attention declaration

Task. A HotpotQA trajectory with its supporting paragraphs scattered across the context is split into $K = 8$ labelled regions, and the model gets a question whose answer lies in exactly one. Rendered contexts reach

8,966 tokens, inside every model’s 32,768-token window. 384 probes; the gold region is permuted per probe; three seeds reseed the layouts and the shuffle over a fixed probe set, and the gold distribution for each seed is pinned so every scale sees identical layouts.

Elicitations.

- **generate** — the model replies **FOCUS**: $\langle k \rangle$, parsed like a tool call, as in Ho et al. (2026).
- **read** — for each region we ask whether it contains the answer and score $\log p(\text{Yes}) - \log p(\text{No})$; the argmax is the choice. K batched region prompts.
- **attention** — no elicitation: we run the model on the full context and question and take the attention mass the final query position places on each region, summed over layers. **attention_late** restricts the sum to the second half of the layers. One forward pass over roughly the same tokens as **read**.

Results (mean $\pm$ sd over three seeds; chance 0.125, best-constant control 0.148):

model	elicitation	hit	σ over constant	content dependence	slot stable	unparsed
0.5B	generate	0.193 ± 0.012	2.2	0.085 ± 0.021	0.281	0.019
0.5B	attention	0.151 ± 0.003	0.2	0.014 ± 0.012	0.882	0.000
0.5B	attention_late	0.234 ± 0.011	4.0	0.113 ± 0.029	0.356	0.000
0.5B	read	0.332 ± 0.004	7.7	0.203 ± 0.003	0.136	0.000
1.5B	generate	0.150 ± 0.008	0.1	0.030 ± 0.023	0.413	0.170
1.5B	attention	0.286 ± 0.023	6.0	0.163 ± 0.010	0.447	0.000
1.5B	attention_late	0.374 ± 0.017	9.2	0.258 ± 0.028	0.200	0.000
1.5B	read	0.378 ± 0.005	9.3	0.241 ± 0.005	0.136	0.000
7B	generate	0.230 ± 0.010	3.8	0.116 ± 0.020	0.260	0.000
7B	attention	0.229 ± 0.009	3.8	0.085 ± 0.018	0.595	0.000
7B	attention_late	0.359 ± 0.017	8.6	0.223 ± 0.019	0.242	0.000
7B	read	0.418 ± 0.011	10.7	0.285 ± 0.004	0.135	0.000

The stated declaration never approaches a measured one. At 0.5B its raw hit rate clears the constant control ($+2.2\sigma$), but the shuffle shows most of that is position: content dependence 0.085, under half of **read**’s. At 1.5B it is indistinguishable from naming one fixed region ($+0.1\sigma$), refuses the **FOCUS**: format on 17% of probes, and its slot-stable rate rises to 0.41. At 7B the refusals disappear and it clears the constant control again ($+3.8\sigma$, between $+3.7\sigma$ and $+4.0\sigma$ on every seed), with content dependence 0.116 — the 0.5B pattern rather than the 1.5B collapse. **read** still beats it by 0.188 in hit rate ($t(2) = 24.0$) and 0.168 in content dependence ($t(2) = 12.3$). The decline from 0.5B to 1.5B does not continue; the gap to the measured routes does.

read wins at 0.5B and 7B and ties at 1.5B. Against the best attention variant, **read** leads at 0.5B on hit rate (0.332 vs 0.234, $t(2) = 24.9$) and content dependence (0.203 vs 0.113, $t(2) = 4.91$). At 1.5B the two tie: 0.378 vs 0.374 ($t(2) = 0.30$) and 0.241 vs 0.258 ($t(2) = -0.85$). At 7B **read** leads again, 0.418 vs 0.359 ($t(2) = 12.0$) and 0.285 vs 0.223 ($t(2) = 4.69$). **read** is the only signal whose content dependence rises at every scale (0.203, 0.241, 0.285); **attention_late** rises to 1.5B and falls back at 7B (0.113, 0.258, 0.223), so attention probing does not keep closing the gap. A single-seed run had shown attention nominally ahead at 1.5B; three seeds showed that was layout noise.

Layer choice decides whether probing works at all. Summed over every layer, attention is useless at 0.5B: content dependence 0.014, the same slot on 88% of probes. Restricted to the late half it clears its control at every scale. Early layers carry position and dominate the sum — one Qwen2.5-1.5B layer-0 query-key product reaches 152,967 against a late-layer peak near 300. A probing baseline reported without this ablation would understate itself by more than an order of magnitude.

4.3 What the two mechanisms share

On both the compaction decision and the attention declaration, the model’s measured judgment is better than its stated one — the generated **FOCUS**: at 1.5B carries no content signal (0.030) while the same model’s logit read carries 0.241 and its late-layer attention 0.258. The failure is not that small models lack the judgment; it is that they express it far less well as structured output on demand than their logits reveal it, and scale to 7B narrows that only partly. Measuring is also a design space rather than a single method: the self-query and attention probing swap between win and tie across scales, and attention works only from late layers. Of the routes we tested, the self-query is the safer default at every scale. The context gap (§5.2) is measured by construction and needs no elicitation at all.

5 Finding 2: salient is not trainable

5.1 Consolidating on the compaction decision

Method. After every fourth compaction event, a sleep phase runs LoRA SFT on the spans the compactor kept, with a reservoir replay buffer across phases. The compactor is Qwen2.5-1.5B with the logit-read backend; the consolidated and evaluated backbone is Qwen2.5-0.5B.

Benchmark. Synthetic agent trajectories of 120 turns, each carrying six planted facts (**auth_header**, **db_engine**, ...) in its first twelve turns and filler after, so compaction evicts some facts. Each fact gets one probe, 288 per corpus over 48 trajectories. A probe is correct if the normalised gold answer or an alias appears in the output, and *evicted* if the answer is absent from the compacted context — the only probes where consolidation can help. Every question names a project unique to its trajectory (§6 explains why this matters).

Arms. (a) uniform replay: the same sleep phases and budget on uniformly sampled spans, kept and dropped alike; (b) reflection, following Dennis et al. (2026); (c) cascading compaction with no weight update; (d) full context, the ceiling; and ours, compaction-supervised.

method (corpus 0, 114 evicted)	all-probe	evicted
no context (floor)	0.000	0.000
(c) cascading, no adapter	0.413	0.000
ours: compaction-supervised	0.451	0.000
ours + mask head	0.323	0.009
(b) reflection	0.389	0.000
(a) uniform replay	0.465	0.123
(d) full context	0.712	—

Uniform replay recovers 14 of 114 evicted facts (+4.0 σ over zero); compaction-gated consolidation recovers none. Across five independently generated corpora:

corpus	evicted	cascading	compaction	uniform
0	114	0.000	0.000	0.123
1	102	0.000	0.029	0.078
2	112	0.000	0.000	0.009
3	98	0.000	0.020	0.061
4	112	0.000	0.036	0.107
mean		0.000	0.017	0.076

Uniform beats the compaction gate on every corpus, paired $t(4) = 3.10$ against a critical value of 2.78. Corpus 0 was the most favourable draw; at three corpora the margin was not yet significant ($t(2) = 1.81$). The gain is specific to evicted facts: all-probe accuracy is level ($t(4) = -0.88$).

Mechanism: coverage. Both adapters fit their targets equally well (validation CE 0.0003 for ours, 0.0005 for uniform); they differ in what they fit. The compactor never keeps `auth_header` and almost never `config_flag` (§4.1), so the gated adapter never sees them. Uniform sampling covers them. A strong salience signal with a blind spot passes the blind spot to the weights.

Remixing dropped spans does not fix it. Replacing 25% or 50% of the gated budget with randomly drawn dropped spans, at a matched total budget, recovers 2 and 1 of 114. The dropped pool is dominated by filler (5,956 filler spans), so a random draw rarely lands on the excluded facts. Drawing round-robin across fact keys instead roughly doubles recovery at 25% (4 of 114) and falls to zero at 50% — one run each, and far short of uniform’s 14. Uniform’s advantage comes from covering every span across 25 phases, not from any single proportional remix.

The same mask succeeds at the opposite task. Following Khodabandehlou & Krishnamachari (2026), a LoRA trained on the same free label to *refuse* when evidence was evicted abstains on 0.728 of evicted probes against 0.000 for the base model (+17.5 σ), cutting hallucination by 72.8%, while refusing only 10.9% of probes whose answer is still present and raising retained accuracy from 0.684 to 0.770. The compaction mask tells a model what was lost: enough to abstain, not enough to recover.

Natural prose. On HotpotQA (one seed, 160 probes, 92 evicted) nothing separates: cascading with no adapter recovers 4 of 92, compaction 3, uniform 5 (+0.7 σ uniform over compaction). Some answers survive in other paragraphs or in the model’s prior, so the no-training floor is not zero. At this size the run neither replicates nor contradicts the synthetic result.

5.2 Gating distillation on the context gap

Scoring. For a skill document d , demonstration query q and response r , form $T = \text{chat}(\text{system}, d \text{ ++ } q)$ and $S = \text{chat}(\text{system}, q)$, run the frozen backbone on $T \text{ ++ } r$ and $S \text{ ++ } r$, and score each response token by

$$\text{kl_i} = \text{KL}(p(\cdot \mid T, r_{<i}) \parallel p(\cdot \mid S, r_{<i}))$$

Tokens group into sentence and code-line spans via the tokenizer’s offset mapping; a span scores the mean of its tokens.

Gating and loss. The top 25% of spans get weight 1 and the rest a floor weight. The teacher is the backbone with its adapter disabled and the document in context; the student is the adapter-enabled model without it; the loss is $\sum w_i \text{KL}_i / \sum w_i$. One model in memory, one adapter per skill category.

Setup. Eight invented tool-use APIs in three categories, each with a `SKILL.md`, nine demonstrations and twelve held-out tasks (96 in all); the APIs are fictional, so the base model cannot know them. Qwen2.5-1.5B-Instruct, 300 steps at learning rate 1e-4. A task passes when every required API string appears in the output. Arms: **uniform**, every token at full weight (S2L-style (Zhang & Qi, 2026)); **random**, a random 25% of spans at full weight and the floor elsewhere; **kl_top**, the KL-selected 25% under the same floor.

The gate carries a signal. Required-token coverage — the share of each demonstration’s required API identifiers inside the selection — is **0.597 against 0.392** for a matched-budget random selection, +6.77 σ ; at token granularity, 0.363 vs 0.247, +4.29 σ .

It does not help training. At floor weight 0, **kl_top** reaches 0.510 against **random**’s 0.583 and **uniform**’s 0.708: the prefix that conditions each selected identifier receives no gradient. A floor of 0.1 repairs this:

seed	kl_top	random	uniform
0	0.729	0.729	0.708
1	0.823	0.750	0.729
2	0.802	0.719	0.698
3	0.813	0.792	0.729
4	0.656	0.729	0.677

seed	kl_top	random	uniform
mean	0.765	0.744	0.708

kl_top and **random** share identical masking, so their difference isolates the ranking: **+0.021**, **t(4) = 0.74**, with the sign flipping between seeds. The ranking adds nothing over random selection. Both masked arms beat full-weight **uniform** (**random** +0.035, $t(4) = 3.90$; **kl_top** +0.056, $t(4) = 2.33$, not significant). That is a separate finding about masked versus full-weight training, and **uniform** is not simply mistuned: it gets monotonically worse from 300 to 1,200 steps (0.708 to 0.656), and the best of five learning rates from $5e-5$ to $8e-4$ reaches 0.750 (single seed each). At token granularity with floor 0.1, **kl_top** and **random** are again level (0.771 vs 0.760). The distilled skill keeps partial function under a mismatched retrieved document (0.292 vs 0.000 for the prompted skill).

5.3 Two failures, two reasons

The compaction gate is *harmful*: its selection systematically excludes whole fact types, and no remixing tested restores them. The context-gap gate is *neutral*: its selection is informative about which tokens matter, but training on it is no better than training on a random subset of the same size. Both signals clear matched controls; neither beats uncurated coverage. A signal’s strength says whether it carries information; it does not say whether that information is what the weights are missing. Before gating training on any free signal, compare it against uniform sampling at a matched budget — the matched control of §6, one level up.

6 Matched controls, and what they caught

A raw signal-strength figure is close to meaningless without a control beside it. We report every one against a matched control because doing so repeatedly caught us:

Artifact	Apparent effect	Caught by
Silent fallback to a rule-based compactor	3.99x lift	decision-provenance counter
“Keep at most N” answered NONE on every event	100% empty keeps	empty-keep rate
Model answered with the first N spans	1.68x lift	positional control
<code>sorted(kept)[:budget]</code> reimposed position after shuffling	1.69x lift	kept-index distribution
Supporting paragraphs stacked at the front of a benchmark	1.76x lift	positional control (2.44x)
Cached scores from an older tokenizer inverted the gate	-5.14σ	re-scoring in the current environment (+6.40 σ)
Falling CE on gold answers read as knowledge transfer	6.29 $\rightarrow$ 2.24	distractor ranking (at chance)
Eval questions that did not identify which item was meant	50% ceiling	duplicate-answer audit

Six of the eight inflated a result in our favour; two hid a possible positive. None was a modelling error; all were construction, which we think is the norm.

The tokenizer row moved the result against us, so disbelief did not catch it, and we misdiagnosed it once. Scored from a cached file produced under an older tokenizer, which split identifiers such as `vx_stash` into five pieces, code spans averaged low and the gate scored 0.118 required-token coverage against a matched control’s 0.307. We blamed mean-pooling. Re-scoring in the current environment, with no code change, gave

0.586 against 0.392. Having seen the stale number, we also marked the downstream comparison confounded; re-running it reproduced the original result exactly. A retraction needs the same evidence as a claim, and a cached score file is valid only in the environment that produced it.

The last two rows compound. Consolidation lowered per-token CE on evicted gold answers from 6.29 to 2.24, which we first read as knowledge that had entered the weights; ranking each gold answer against same-bank distractors put every arm at the 0.240 chance rate, which we then read as knowledge never acquired. Both were premature. 98.3% of the benchmark’s questions had more than one correct answer across trajectories — “Which header carries the auth token?” was asked of 48 conversations with four different answers — so a perfect memoriser could not exceed 50%, and the distractors were other trajectories’ correct answers. Scoping every question to its trajectory is what made §5.1 measurable. The audit — count the distinct answers each question string receives — costs minutes and should precede the experiment.

7 Discussion

The practical rule for free-supervision pipelines below frontier scale is to **derive the signal from what the model does, not from what it says it does**, and then to check the signal against uniform sampling before training on it. The context gap obeys the first half by construction; the compaction decision obeys it once read rather than generated; Declarative Attention as published does not, and §4.2 measures the cost at small scale. The second half is where both consolidation signals fail.

The two halves interact. The measured compaction decision is strong precisely because it reflects what the compactor considers salient — and that is also why it fails as a training target, since what a compactor keeps for immediate use need not be what the weights lack. A signal that is maximally informative about one decision can be systematically biased for another. This suggests the useful free signals for consolidation are ones whose errors are uncorrelated with coverage, or ones used, as in abstention, for the decision they were made for.

For routing, the logit read is a cheap, training-free substitute for declaring at the scales where declaring is unreliable. It is not a deployable router on its own — about 3x chance leaves most probes misrouted — and it costs K region prompts rather than a single generated token. Whether it keeps its lead where declaration works, at 27B and above, is the natural next test.

8 Limitations

- **Synthetic consolidation.** The significant consolidation result is on a controlled synthetic benchmark. The HotpotQA run is one seed at 92 evicted probes with a non-zero no-training floor and cannot resolve a gap of the synthetic size. The synthetic generator’s unmarked variant is a floor case: facts and filler share one template bank.
- **Seeds.** The main comparisons have five seeds (§5.1 corpora, §5.2 gate) or three (§4.2); the remix ablations, abstention, the step and learning-rate sweeps, and HotpotQA are single runs.
- **Scale.** Everything is at or below 7B, and 7B is 4-bit quantised. We cannot test the 27B+ regime of Ho et al. (2026), so our result bounds the generated declaration from below — at a fixed-slot control at 1.5B, under half the logit read’s signal at 7B — and does not contradict the accuracy Ho et al. (2026) reports at scale. The logit read is untested there.
- **Routing accuracy.** The best router is about 3x chance; §4.2 compares elicitations, it does not deliver a usable router.
- **One model family.** Apart from SmoLM2-360M in §4.1, all results are on Qwen2.5.

9 Conclusion

Free salience signals exist at small scale, but only when measured: asked to state what matters, 0.5B–7B models fail or carry a fraction of the signal their logits hold. And a measured signal that clears its control can still be the wrong thing to train on — compaction-gated consolidation loses to uniform replay on every

corpus, and context-gap gating adds nothing over random selection. Both lessons reduce to the same practice: put a matched control beside every number, including the number that says a signal should be trained on.

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A Hyperparameters

	§5.1 consolidation	§5.2 distillation	§4.2 routing
backbone	Qwen2.5-0.5B (compactor 1.5B)	Qwen2.5-1.5B	Qwen2.5-0.5B / 1.5B / 7B
LoRA	r 16, α 32, dropout 0.05, all projections	same	—
learning rate	5e-5	1e-4	—
steps	80 per sleep phase	300	—
schedule	sleep every 4 compaction events, 25 phases	one adapter per category	—
replay	reservoir, capacity 80	—	—
selection	keep $\rho = 0.25$, sentence spans, 8 recent turns protected	top 25% of spans, floor 0 or 0.1	K = 8 regions
data	48 trajectories $\times$ 120 turns $\times$ 6 facts	8 APIs, 9 demos, 12 tasks each	96 HotpotQA trajectories $\times$ 4 probes
seeds	5 corpora	5	3 layouts
precision	fp16	fp16	fp16; 7B nf4

B Reproducibility

All code, configs and Kaggle notebooks are in the supplementary repository. Each result above maps to one config and one notebook; `docs/project_a_findings.md`, `docs/project_b_findings.md` and `docs/declarative.md` hold the per-run tables, and `eval/declare_report.py` regenerates the §4.2 table and paired tests from the run directories. The 7B routing runs use `configs/kaggle_declare_7b.yaml` with a repeat-KV SDPA attention (`declare/elicite.py`) that matches stock SDPA exactly but avoids a 5 GB attention matrix on T4 hardware for 9k-token prompts.